# MAREESHWARAN S

**ECE Student | Embedded Systems & IoT (ESP32, Arduino)**

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## PROFESSIONAL SUMMARY

Electronics and Communication Engineering student (2024–2028) who builds working hardware, not just circuits on paper. Hands-on with ESP32 and Arduino, Bluetooth and Wi-Fi control, motor drivers, OLED/LCD displays and RTC modules across six projects, plus an internship at Green Microelectronics India Pvt. Ltd. Seeking an embedded systems or electronics internship to apply C programming and prototyping skills on real products.

## SKILLS

- **Programming:** C (learning), Embedded C basics
- **Microcontrollers:** ESP32, Arduino Uno
- **Industrial automation:** PLC design, timer circuits, tower lamp circuits
- **Soft skills:** Teamwork, presentation, quiz and ideation competitions

## EXPERIENCE

**Intern | Green Microelectronics India Pvt. Ltd., Coimbatore**

- Worked on PLC design and control logic.
- Designed a timer circuit.
- Designed a tower lamp circuit.

## PROJECTS

**1. EMO AI Assistant Mini Robot** | ESP32 | OLED | DS3231 RTC | Touch Sensor | Mobile App

- Designed and built an interactive mini robot on ESP32 with animated facial expressions shown on an OLED display.
- Integrated a DS3231 real-time clock over I2C to add real-time clock functionality to the robot.
- Built a mobile-based question-and-answer feature that sends text replies to the OLED screen.
- Added a touch sensor that changes the robot's expression on tap, making the interaction feel natural.
- Tested each module separately: detected the OLED (0x3C) and RTC (0x68) on the I2C bus, and checked RTC time and touch readings in the Serial Monitor.

**2. ESP32 Three-Wheel Wi-Fi Rover** | ESP32 | Motor Driver | Wi-Fi | Power Management

- Designed and assembled a three-wheel rover with a motor driver, speed motors and a chassis.
- Developed Wi-Fi-based remote control from a mobile device for smooth movement and navigation.
- Handled motor control and battery power management on the rover.

**3. Smart Mobile-Controlled Robot (Team Project)** | ESP32 | Bluetooth | LCD | 12V Battery

- Developed an ESP32-based rover using torque motors and a 12V battery, with an LCD showing status.
- Worked with the team on robot assembly and on the Bluetooth control features.

**4. Arduino Bluetooth RC Car** | Arduino Uno | HC-05 | Motor Driver | 12V Li-ion

- Built a Bluetooth-controlled car using Arduino Uno, an HC-05 module, a motor driver and DC motors.
- Enabled wireless driving from a smartphone app, powered by a 12V Li-ion battery.

**5. Mini Quadcopter Drone** | Motors | Flight Controller | Battery | Propellers

- Assembled and tested a mini quadcopter by integrating motors, a flight controller, battery and propellers.
- Gained hands-on experience in drone assembly, testing and electronics troubleshooting.

**6. Li-Fi Text Transmission Prototype** | ESP32 | ESP8266 | LED | Solar Panel | C++

- Built a Li-Fi link that sends text from one laptop to another using light: an ESP32 drives an LED and a solar panel with an ESP8266 receives it.
- Wrote a C++ program (compiled with g++ in MSYS2) that reads a text file and sends its contents; tested up to 500 characters.
- Tested at 2 cm in a closed setup, one-way. Planned next steps: two-way link, faster photodiode receiver, and file transfer with error checking.

## EDUCATION

**B.E. Electronics and Communication Engineering** | 2024 – 2028 | CGPA: 7.9 / 10
Tamil Nadu College of Engineering, Coimbatore (Affiliated to Anna University, Chennai)

**Higher Secondary (HSC)** | 2024 | 78.67% (472 / 600)
SKA Higher Secondary School, Aundipatti, Tamil Nadu

## PUBLICATION

- **Preprint, Zenodo:** "OLED EMO Robot: A Low-Cost ESP32-Based Expressive Clock with Touch Interaction and a Planned Offline Voice Interface". Sole author. DOI: 10.5281/zenodo.23387877
- **Preprint, Zenodo:** "Design of a Low-Cost Li-Fi Text Transmission Prototype Using an LED and Solar-Panel Receiver, with a Proposed Extension to Files, Images, Audio and Video". Sole author. DOI: 10.5281/zenodo.23389126

## ARTICLES & PROJECT PAGES

- **Medium:** "EMO Mini Assistant: An ESP32-Based Emotion-Interactive Companion".
- **Hackster.io:** "Visible Light Communication (Li-Fi) for Text Data Transmission".

## ACHIEVEMENTS & AWARDS

- **2nd Prize, Ideathon (Li-Fi Technology):** Rathinam Group of Institutions.
- **3rd Prize, Quiz Competition:** CSI College of Engineering.
- **3rd Prize, PPT Presentation:** P.A. College of Engineering and Technology.